

Facilities Study Memorandum

For Physical Interconnection of a Large Customer Load

Project: Prairie Horizon Data Campus

Project number: PHDC-LL-2026-017

Point of interconnection: new customer breaker-and-a-half bay at Jefferson 138 kV Station

Date: April 2026

Introduction

This memorandum has been prepared to support preliminary review of the physical interconnection requirements for Prairie Horizon Data Campus, a large customer load proposed in Fayette County, Ohio. The customer proposes a phased campus load reaching an ultimate maximum demand of 180 MW at the 138 kV point of interconnection. The transmission owner is AEP Ohio Transmission Company, Inc. and the transmission provider is PJM Interconnection, L.L.C.

A. Project Description

The customer proposes six data halls and associated utility, cooling, and administration facilities on approximately 212 acres. The campus medium-voltage system is 13.8 kV. Main transformation is provided by three customer-owned power transformers rated 100/133/167 MVA each. Ultimate campus loading is phased as 60 MW, 120 MW, and 180 MW. The customer represents that standby generation is emergency-use only and is not intended to provide wholesale or retail market service across the point of interconnection.

B. Point of Interconnection and Ownership Demarcation

The proposed point of interconnection is the new customer breaker-and-a-half bay at Jefferson 138 kV Station. The customer-owned high-side facilities include line-terminal disconnects, breakers as assigned by the final substation configuration, metering interface equipment, relay panels, dead-end structures, and the customer transformer bank. The point of change in ownership is the first dead-end structure inside the customer fence line on the customer side of the 138 kV line terminal.

C. Preliminary Scope of Customer Interconnection Facilities

- 138 kV station integration - One new customer 138 kV bay integrated into the existing Jefferson 138 kV Station arrangement.
- Main transformation - Three customer-owned 138/13.8 kV power transformers with associated foundations, oil containment, firewalls, and arresters.
- 13.8 kV main distribution - Main switchgear building with sectionalized buses, feeder breakers, capacitor-bank feeders, and relay/control interfaces.
- Protection and controls - Relay house, metering interface cabinet, telecom demarcation, station batteries, and AC/DC auxiliary systems.
- Civil works - Access-road improvements, grounding-grid extension, fiber duct bank, and control-cable raceways.

D. Transmission Owner / Utility Review Topics

- Short-circuit duty - Final breaker duties and CT requirements to be confirmed by TO studies.
- Voltage performance - Customer will maintain 0.98 lagging or better at the POI using staged capacitor banks.
- Protection integration - Final relay philosophy and communications-assisted trip scheme subject to TO protection review.
- Metering - Revenue metering at the 138 kV high side using a TO-approved PT/CT arrangement.
- Operations - No normal export and no unreviewed parallel operation of emergency generation across the POI.
- Construction sequencing - T1 / Bus A energized first, followed by T2 / Bus B, then T3 / Bus C.

E. Preliminary Cost and Schedule Considerations

Station integration scope remains subject to transmission-owner scope definition, outage windows, protection changes, and station standards. The customer substation is customer-provided under EPC contract. Telecommunications and SCADA integration remain dependent on TO and utility communications architecture and control-center requirements.

F. Remaining Engineering Deliverables

The following items remain to be completed during detailed design or subsequent study stages: detailed short-circuit calculations, harmonic analysis, grounding study, final relay-setting calculations, AC/DC elementary diagrams, panel layouts, conduit and cable schedules, and final station integration drawings. This memorandum is intended only to provide a coordinated engineering narrative for interconnection review.